Huawei Access Node - Optical Transport Failure Resolution Procedure for Node ID: JAY-ACC-39 at Jayanagar Metro Site

Node: JAY-ACC-39 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_3652

Description

Optical Transport Failures on Huawei Access Nodes (Role: Access) at the Jayanagar Metro Site are typically caused by SFP/XFP degradation or fiber cuts, resulting in a loss of receive (Rx) or transmit (Tx) power, ultimately triggering the Major Alarm Code: ALARM_3652. This failure domain occurs due to the inherent mechanical and optical limitations of the SFP/XFP modules, which can be prone to degradation over time or damaged by physical stress. Furthermore, the use of optical fibers as a medium for signal transmission introduces an additional layer of vulnerability, as fiber cuts or breaks can disrupt the signal path. To mitigate this risk, Huawei's Access Nodes employ a robust optical transport mechanism, which includes redundant links and automatic protection switching (APS) to ensure high availability. However, in the event of an Optical Transport Failure, it is essential to promptly diagnose and resolve the issue to minimize downtime and ensure uninterrupted service.

Pre-requisite Checks

- Verify that the site location is Jayanagar Metro Site and the affected Node ID is JAY-ACC-39.

- Confirm that the Role of the affected node is Access and the Vendor is Huawei.

- Check if the redundant link is operational and available for switchover.

- Ensure that the necessary permissions and access rights have been granted to perform the diagnostic and resolution commands.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-fiber link-status, Expected Output: The display will show the status of the optical fibers, including the link status, power level, and signal quality., Description: This command will isolate the issue by displaying the link status, power level, and signal quality of the optical fibers, helping to determine if the issue is related to a fiber cut or degradation.
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```
Command: display optical-fiber sfp- status, Expected Output: The display will show the SFP status, including the SFP type, status, and alarm information., Description: This command will isolate the issue by displaying the SFP status, including the SFP type, status, and alarm information, which can help identify if the issue is related to an SFP/XFP degradation or failure.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module, Command: replace optical-fiber sfp [slot-number], Description: This step involves replacing the faulty SFP/XFP module with a new one to restore the optical transport functionality.

Step 2: Step: Repair or replace the damaged fiber, Command: repair optical-fiber [fiber-id], Description: This step involves repairing or replacing the damaged fiber to restore the optical transport functionality.

Post-Check Verification

```
Command: display optical-fiber link-status, Expected Output: The display will show the status of the optical fibers, including the link status, power level, and signal quality., Description: This command will verify if the issue has been resolved by displaying the link status, power level, and signal quality of the optical fibers.

Command: display optical-fiber sfp-status, Expected Output: The display will show the SFP status, including the SFP type, status, and alarm information., Description: This command will verify if the issue has been resolved by displaying the SFP status, including the SFP type, status, and alarm information.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: undo replace optical-fiber sfp [slot-number], Description: This step involves restoring the original SFP/XFP module to its previous configuration.

- Step: Restore the original fiber configuration, Command: undo repair optical-fiber [fiber-id], Description: This step involves restoring the original fiber configuration to its previous state.